1 Integer ambiguity resolution

Integer ambiguity resolution (AR) maps the real-valued, bias-contaminated phase ambiguities of the float solution to the integer carrier-cycle counts they represent. The engine runs MLAMBDA [3] on a between-satellite-differenced ambiguity sub-problem extracted from the filter state and covariance. The operative quantity is the ambiguity differenced against a per-constellation pivot p_g (the differencing section of the Foundations document). The single-receiver and baseline modes share the integer-estimation machinery, but the condition that makes the differenced ambiguity an integer differs by mode.

The undifferenced ambiguity $N^i_{r,s}$ of the phase observation equation (Foundations, observation equations) carries a receiver phase bias $b_{r,s}$ and a satellite phase bias b^i_s. In single-receiver PPP and PPP-RTK the pivot difference is a *between-satellite single difference* $\Delta\,\cdot$ (the between-satellite single-difference section of the Foundations document). It cancels the receiver bias $b_{r,s}$, which is common to every tracked satellite, and leaves the satellite bias. The residual $\Delta N^{i,p_g}_{r,s}$ is therefore integer only after b^i_s is removed by a consistent external phase-bias product (BIA or SSR, the SSR-corrections document). In baseline mode the state holds true double differences $\nabla\Delta\,\cdot$ (the double-difference section of the Foundations document); the additional between-receiver difference also cancels the satellite bias, so $\nabla\Delta N$ is integer without an external product. The equations below keep the resolver's internal label $\nabla\Delta$ for the differenced ambiguity, since the algebra is the same in both cases: it denotes the between-satellite single difference Δ in the single-receiver case and the full double difference only in baseline mode.

The implementation separates a driver from a numerical kernel. The driver, `src/core/filter/AmbiguityResolver.cpp`, assembles the state sub-vector, rescales between metres and cycles (metre↔cycle), looks up the per-pair λ, forms the differences, performs the WL/NL decomposition and applies the returned integer fix. The numerical core lives in the `GnssProcessor::ils` namespace of `src/core/filter/IlsCore.{hpp,cpp}`: the LDL^{T} factorisation (in the factor order of (5)), the $\boldsymbol{Z}$-reduction, the Schnorr–Euchner search, the bootstrap success rate and the full acceptance battery. These are pure free functions over ($\hat{\boldsymbol{N}}$ cycles, $\boldsymbol{Q}_{\hat{N}}$ cycles^2), with no dependence on `StateManager`, `KalmanFilter` or the configuration. The PPP-RTK network producer (the Network mode document) calls the same `ils` core under its own gate set, as a read-only fixability probe only, because the producer is float-only (Section 1.7). The single-receiver and baseline paths below are configured under the JSON path `estimation.ambiguity_resolution.*`.

1.1 The integer least-squares problem

The float solution gives a real-valued ambiguity estimate $\hat{\boldsymbol{N}}$ (in cycles) with covariance $\boldsymbol{Q}_{\hat{N}}$, both extracted from the filter state $\boldsymbol{x}$ and covariance $\boldsymbol{P}$. Integer ambiguity resolution is the weighted closest-lattice-point problem

$$\boxed{\check{\boldsymbol{N}} \;=\; \arg\min_{\boldsymbol{a}\,\in\,\mathbb{Z}^n} \big(\hat{\boldsymbol{N}}-\boldsymbol{a}\big)^{\mathrm{T}}\,\boldsymbol{Q}_{\hat{N}}^{-1}\,\big(\hat{\boldsymbol{N}}-\boldsymbol{a}\big),} \tag{1}$$

that is, the minimisation of the squared Mahalanobis distance from $\hat{\boldsymbol{N}}$ to a lattice point $\boldsymbol{a}$ in the metric of $\boldsymbol{Q}_{\hat{N}}^{-1}$. This integer least-squares (ILS) problem is NP-hard. LAMBDA and its MLAMBDA variant make it tractable in three stages, none of which has a tuning parameter: (i) an LDL^{T} factorisation of $\boldsymbol{Q}_{\hat{N}}$, (ii) a decorrelating integer $\boldsymbol{Z}$-transform (lattice reduction), and (iii)

a depth-first shrinking-ellipsoid search over the reduced lattice. The hat/check convention of the notation block applies: $\hat{\boldsymbol{N}}$ is the float estimate and $\check{\boldsymbol{N}}$ the integer-fixed solution.

Float extraction and rescaling to cycles. The filter stores per-band ambiguity states in metres ($\lambda_s N$, one per tracked satellite and signal). The resolver therefore forms the between-satellite-differenced float and covariance by explicit differencing against the pivot p_g, then rescales by wavelength. In baseline mode the state already holds the double difference, and the differencing step reduces to a direct selection (Section 1.6). For a differenced pair k with satellite index $i(k)$ and pivot index $p(k)$ (`src/core/filter/AmbiguityResolver.cpp`):

$$y_k^{\nabla\Delta} = \boldsymbol{x}_{\,i(k)} - \boldsymbol{x}_{\,p(k)}, \qquad \text{(DD float, m)} \tag{2}$$

$$\boldsymbol{Q}_{kl}^{\nabla\Delta} = \boldsymbol{P}_{i_k i_l} - \boldsymbol{P}_{i_k p_l} - \boldsymbol{P}_{p_k i_l} + \boldsymbol{P}_{p_k p_l}, \qquad \text{(DD covariance, m}^2\text{)} \tag{3}$$

$$\hat{\boldsymbol{N}}_k = y_k^{\nabla\Delta}/\lambda_k, \qquad (\boldsymbol{Q}_{\hat{N}})_{kl} = \boldsymbol{Q}_{kl}^{\nabla\Delta}/(\lambda_k\lambda_l). \qquad \text{(cycle domain)} \tag{4}$$

The rescaling (4) is required. On metre-domain floats MLAMBDA would round toward integer-metre values, which have no carrier-cycle meaning, and the ratio test (Section 1.5) would return ≈ 1 irrespective of float quality. The per-pair wavelength λ_k uses the GLONASS FDMA per-satellite frequency where applicable. $\boldsymbol{Q}^{\nabla\Delta}$ is symmetrised, $\boldsymbol{Q}^{\nabla\Delta} \leftarrow \frac{1}{2}(\boldsymbol{Q}^{\nabla\Delta} + \boldsymbol{Q}^{\nabla\Delta\mathrm{T}})$, before factorisation.

The resolver elects its own DD reference: in each (constellation, signal) group of AR-eligible ambiguities, the highest-elevation member is the reference and every other member is differenced against it. This reference differs from the filter's BSSD pivot. Under BSSD the pivot's ambiguity is frozen at zero variance, which excludes it from the candidate set (Section 1.4); the resolver therefore differences against a second reference, and the resulting quantity is a difference of single differences. The two elections also use different rules. The BSSD pivot is chosen on a hard phase-coverage tier, a convergence/elevation score and an ephemeris-freshness bar (the differencing section of the Foundations document); the AR reference is chosen on elevation alone (`AmbiguityResolver::buildDdPairs`).

1.2 Decorrelation: LDL^{T} factorisation and the integer Z-transform

Factorisation. MLAMBDA factorises the symmetric positive-definite ambiguity covariance as

$$\boldsymbol{Q}_{\hat{N}} \;=\; \boldsymbol{L}^{\mathrm{T}}\,\boldsymbol{D}\,\boldsymbol{L}, \qquad \boldsymbol{L} \text{ unit lower-triangular}, \;\; \boldsymbol{D} = \mathrm{diag}(d_1, \ldots, d_n), \;\; d_i > 0, \tag{5}$$

eliminated in reverse index order, from the last row up (the Chang–Yang MLAMBDA convention). The transposed unit lower-triangular factor stands on the left, as written in (5); the shorthand "LDL-transpose" used in this section names that factorisation, not the reversed factor order. A non-positive pivot $d_i \le 0$ signals a covariance that is not positive definite and aborts the fix. After reduction, d_i is the *conditional variance* of the i-th decorrelated ambiguity given the later ones; the success-rate gate of Section 1.5 uses it directly.

The Z-transform. The central device of LAMBDA [3] is a unimodular integer transform $\boldsymbol{Z} \in \mathbb{Z}^{n\times n}$, $|\det \boldsymbol{Z}| = 1$, that re-parameterises

$$\boldsymbol{z} = \boldsymbol{Z}^{\mathrm{T}}\hat{\boldsymbol{N}}, \qquad \boldsymbol{Q}_{\hat{z}} = \boldsymbol{Z}^{\mathrm{T}}\boldsymbol{Q}_{\hat{N}}\,\boldsymbol{Z}, \tag{6}$$

so that the transformed ambiguities are nearly decorrelated (small off-diagonal $\boldsymbol{L}$ entries) and their conditional variances are ordered. The decorrelated float vector is $\bar{\boldsymbol{z}} := \boldsymbol{Z}^{\mathrm{T}}\hat{\boldsymbol{N}}$, with components $\bar{z}_k$;

it is the quantity the search of Section 1.2 rounds, computed as `z_float = Z.transpose()*a` in `ils::mlambda` (`src/core/filter/IlsCore.cpp`). Because $\boldsymbol{Z}$ is unimodular, $(\boldsymbol{Z}^{\mathrm{T}})^{-1}$ is also integer and the integer minimiser maps back exactly, $\check{\boldsymbol{N}} = (\boldsymbol{Z}^{\mathrm{T}})^{-1}\check{\boldsymbol{z}}$. Decorrelation therefore leaves the solution of (1) unchanged. It makes the search ellipsoid closer to spherical, which reduces the number of enumerated points by orders of magnitude. The reduction interleaves two operations in a sweep from $j = n - 2$ downward.

Integer Gauss transformations reduce the sub-diagonal $\boldsymbol{L}$ entries by integer column operations,

$$\mu = \lceil L_{ij} \rfloor, \qquad \boldsymbol{L}_{(:,j)} \mathrel{-}= \mu\, \boldsymbol{L}_{(:,i)}, \qquad \boldsymbol{Z}_{(:,j)} \mathrel{-}= \mu\, \boldsymbol{Z}_{(:,i)}, \tag{7}$$

where $\lceil \cdot \rfloor$ is round-to-nearest, keeping every $|L_{ij}| \le \frac{1}{2}$.

Permutations (conditional adjacent swaps) reorder two ambiguities when the swap reduces the later conditional variance. With $\bar{d}_j = d_j + L_{j+1,j}^2\, d_{j+1}$, the swap fires when

$$\bar{d}_j + \varepsilon \;<\; d_{j+1}, \qquad \varepsilon = 10^{-6}, \tag{8}$$

after which d_j, d_{j+1}, the affected rows/columns of $\boldsymbol{L}$ and the columns of $\boldsymbol{Z}$ are updated by the standard rank-one formulas ($\eta = d_j/\bar{d}_j$, $\lambda = d_{j+1}L_{j+1,j}/\bar{d}_j$, $d_j \leftarrow \eta\, d_{j+1}$, $d_{j+1} \leftarrow \bar{d}_j$), and the sweep restarts from $j = n - 2$. The guard $\varepsilon = 10^{-6}$ prevents repeated swaps of near-equal variances. It is a numerical safeguard rather than an estimator setting and has no configuration key.

The shrinking-ellipsoid search. The reduced problem is solved by depth-first enumeration of the lattice points inside the ellipsoid $\sum_i (z_i - \hat{z}_i)^2/d_i \le \chi^2$, in Schnorr–Euchner order (nearest integer first, then zig-zagging outward). Recursing from the last level $k = n - 1$ down to 0,

$$\hat{z}_k = \bar{z}_k + \sum_{i>k} (z_i - \hat{z}_i)\, L_{ik}, \qquad \text{(conditional centre)} \tag{9}$$

$$z_k = \lceil \hat{z}_k \rfloor, \qquad \mathrm{dist}_k = \mathrm{dist}_{k+1} + \frac{(\hat{z}_k - z_k)^2}{d_k}. \qquad \text{(partial distance)} \tag{10}$$

A complete integer vector that reaches level 0 with cumulative distance below the current bound χ^2 is recorded. Once the candidate list is full, the bound shrinks to the worst retained candidate, so the ellipsoid contracts as better candidates are found. The search (`ils::search` in `src/core/filter/IlsCore.cpp`) retains the $m = 2$ best candidates for the ratio test, sorted by ascending squared distance, under a hard iteration cap (`kMaxIter` = 10 000). Fewer than two candidates is a failure. The outputs are the two smallest objective values $s_1 \le s_2$ of (1) and the best integer vector $\check{\boldsymbol{z}}$, back-transformed via $\check{\boldsymbol{N}} = (\boldsymbol{Z}^{\mathrm{T}})^{-1}\check{\boldsymbol{z}}$.

1.3 Wide-lane / narrow-lane decomposition (PPP path)

PPP and PPP-RTK AR use a *wide-lane-first* decomposition when Melbourne–Wübbena (MW) resolved wide-lane (WL) integers are available. It halves the search dimension and exploits the long WL wavelength [2]. The driver attempts the WL-constrained narrow-lane (NL) fix before any unconstrained LAMBDA. When WL integers exist and the constrained fix fails, the epoch stays float: the driver does not fall through to an unconstrained search, because that search would include DDs whose WL is not locked and could return integers that contradict the MW wide lane. A wrong-WL fix is worse than no fix, so the enforced rule is "no NL AR for satellites whose WL is float" (`src/core/filter/AmbiguityResolver.cpp`).

WL from Melbourne–Wübbena (integer by contract). The DD wide-lane integer is not searched. It is the difference of the MW detector's single-difference WL integers (`src/core/slip/MwSlipDetector`):

$$\nabla\Delta N_{\mathrm{WL}} = N^{i}_{\mathrm{WL}} - N^{p_g}_{\mathrm{WL}} \in \mathbb{Z}, \qquad \lambda_{\mathrm{WL}} = \frac{c}{f_1 - f_2}. \tag{11}$$

NL search variable. For each dual-frequency DD pair the transform $\boldsymbol{T}$ forms the float narrow-lane ambiguity in NL cycles. The default row is the narrow-lane combination below. It is not ionosphere-free: its slant-ionosphere sensitivity is 0.959 NL cycles per TECU, so the default per-pair cap (17) of 0.15 cycles implicitly requires the slant ionosphere to about 0.156 TECU.

$$\hat{n}_{\mathrm{NL}} = \frac{f_1\,\nabla\Delta N_1 + f_2\,\nabla\Delta N_2}{f_1 + f_2} = \frac{1}{f_1 + f_2}\left(\frac{f_1}{\lambda_1}\,\nabla\Delta_1 + \frac{f_2}{\lambda_2}\,\nabla\Delta_2\right), \qquad \lambda_{\mathrm{NL}} = \frac{c}{f_1 + f_2}, \tag{12}$$

with reduced covariance $\boldsymbol{Q}_{\mathrm{NL}} = \boldsymbol{T}\,\boldsymbol{Q}^{\nabla\Delta}\,\boldsymbol{T}^{\mathrm{T}}$. Only NL-paired DDs enter LAMBDA; unpaired single-signal DDs are excluded so that they cannot distort the ratio test. An optional ionosphere-free search row (configuration key nl_iono_free_row, default off) targets the same L1 integer with zero slant-ionosphere sensitivity. For GPS L1/L2 it multiplies the WL sensitivity by 8.06 (a 1-cycle WL error moves the L1 integer target by 3.53 cycles instead of 0.438) and the searched variance by about 65. The equations of this subsection describe the default narrow-lane row.

Integer-recentring shift. The float $\hat{n}_{\mathrm{NL}}$ is not integer-valued in general. Substituting $N_2 = N_1 - \mathrm{WL}$ gives $\hat{n}_{\mathrm{NL}} = N_1 - \frac{f_2}{f_1+f_2}\,\mathrm{WL}$, which for GPS L1/L2 with $\mathrm{WL} = 1$ lies at $N_1 - 0.4379$. Neither neighbouring integer is then materially closer, and the ratio test settles near 1.6 regardless of float quality. The resolver therefore adds the known-WL offset, which makes the search variable integer in expectation,

$$\tilde{n}_k = \hat{n}_{\mathrm{NL},k} + \frac{f_2}{f_1 + f_2}\,\nabla\Delta N_{\mathrm{WL},k} \;\approx\; N_{1,k} \;\in\; \mathbb{Z} + \text{noise}, \tag{13}$$

runs MLAMBDA on $\tilde{\boldsymbol{n}}$ (a constant per-row shift leaves $\boldsymbol{Q}_{\mathrm{NL}}$ unchanged), then back-shifts $\check{n}_{\mathrm{NL}} = \check{N}_1 - \frac{f_2}{f_1+f_2}\,\mathrm{WL}$.

Back-transform to per-signal DDs. The fixed integers map back to the two per-signal DDs in metres,

$$\nabla\Delta_1^{\mathrm{fix}} = \left(\check{n}_{\mathrm{NL}} + \tfrac{f_2}{f_1+f_2}\,\nabla\Delta N_{\mathrm{WL}}\right)\lambda_1, \qquad \nabla\Delta_2^{\mathrm{fix}} = \left(\check{n}_{\mathrm{NL}} - \tfrac{f_1}{f_1+f_2}\,\nabla\Delta N_{\mathrm{WL}}\right)\lambda_2, \tag{14}$$

and re-enter the filter through the constrained update of Section 1.6.

State-basis variants. When `estimation.state_ambiguity_basis = nl_wl` the state holds NL/WL ambiguities directly, and there are no per-band DD pairs to enumerate. In this basis only the MW-resolved WL integers are applied, as constraint rows; a full NL LAMBDA search with the WL pre-constrained is not implemented. The complementary WL-only mode (`estimation.ambiguity_resolution.wl_only`, in the `per_band` basis) applies the cycle-domain constraint $\nabla\Delta_1/\lambda_1 - \nabla\Delta_2/\lambda_2 = \nabla\Delta N_{\mathrm{WL}}$ as one pseudo-observation per pair, without an NL search. When MW re-locks at a new integer, the covariance of the affected single-difference states is inflated so that the state moves to the new integer within one epoch.

Both WL-constraint paths (this one and the `nl_wl`-basis constraint) take their pseudo-observation noise from a separate setting, $\boldsymbol{R} = \sigma^2_{\mathrm{wl}}\,\boldsymbol{I}$ with $\sigma^2_{\mathrm{wl}} =$ `wl_only_pseudo_var` (10^{-4} cycles^2, in the cycle

domain), rather than the metre-domain σ^2_{hold} of the NL/DD feedback in (19). Both paths apply three acceptance gates to the MW integers. The first is the per-arc fractional-residual point test (`wl_max_frac`, default 0.40; on both arcs in the `per_band wl_only` path, on the single arc in the NL/WL basis). The second is a bootstrap success-rate gate (key wl_only_ffr, default on): the product of per-arc rounding success rates, taken over distinct arcs, must reach the threshold min_success_rate of (16), and the weakest arc is dropped until it does. The third is an innovation screen (key wl_only_max_innov_cyc, default 0.5 cycle) that rejects a constraint whose asserted WL integer differs from the predicted DD wide lane by more than that amount. The cross-frequency parity check runs only in the `nl_wl` basis (`applyMwWlConstraintsNlWl_` in `src/core/filter/AmbiguityResolver.cpp`). For a triple-frequency satellite it rejects all three WL DDs when the telescoping canonical cycle counts are inconsistent, $\nabla\Delta_{ab} - \nabla\Delta_{ac} + \nabla\Delta_{bc} \neq 0$. The `per_band wl_only` path has no cross-frequency consistency check.

1.4 Partial / subset ambiguity resolution

Partial AR operates at two levels: a static candidate gate that decides which ambiguities enter the integer problem, and an optional backward-elimination search that looks for a fixable subset when the full set fails validation.

Upstream candidate gating (always on). Subset selection begins when the DD pairs are built (`src/core/filter/AmbiguityResolver.cpp`). A satellite/signal ambiguity enters the integer set only if it passes every filter:

- **Initialised**: the state entry is non-zero (seeded by data), and **non-frozen**: its variance is non-zero, which excludes a BSSD-frozen pivot.
- **Elevation**: $\theta_r^i \geq \theta_{\text{AR}}$ (`elevation_mask`). In the single-receiver path the elevation comes from the epoch's observations; an ambiguity whose satellite has no observation is given elevation 0 and is removed by any positive mask (the consequence for fix-and-hold is stated in Section 1.6).
- **Resolvable**: the ambiguity is not flagged unresolvable for lack of bias or antenna (BIA, ATX) coverage. This applies to the single-receiver path; in baseline DD the analogous filter is the `exclude_keys` set of non-integer-capable DDs from `MeasurementModel`.
- **Constellation**: $g(i) \in$ `active_systems`, when that list is non-empty.
- **Pairable**: the (constellation, signal) group holds ≥ 2 satellites, so a pivot difference exists.
- **Mature** (`min_arc_info`, default 0 = off): both constituent arcs of a candidate (the higher-frequency arcs of the satellite and of the reference) have accumulated at least $\sum_k \sin^2 \theta^i_{r,k}$ of elevation-weighted code information since arc birth, in epoch-equivalents at zenith. Maturity is measured in information rather than epochs: under the elevation stochastic model an arc rising off the mask and one at zenith accumulate information at very different rates, so two arcs of equal age are not equally determined. The counter restarts on both ways an arc can end: the key leaving the state vector, and a refloat in place. A refloat keeps the key and would otherwise let the new arc inherit the pre-slip age (`AmbiguityResolver::refreshArcAges_`, keyed on `StateManager::isAmbiguityPendingReseed`). The gate applies only to the WL-constrained NL candidate set of Section 1.3, not to the baseline DD, WL-only or unconstrained full-set paths. It guards against wrong fixes, since one immature candidate can fail the joint ratio test of (15) for the whole set. It does not speed convergence: it admits a candidate only once the candidate is well determined, which is when the float solution gains least from a fix.

The constellation filter keeps a system in the float solution while excluding it from the candidate set. It serves a constellation whose residual model bias would lower the ratio test.

Dynamic backward-elimination subset (`enable_partial_ar`, default off). With the candidate set fixed, the default is all-or-nothing: the full set is searched and either passes all gates of Section 1.5 or the epoch stays float. When `enable_partial_ar` is set, a failure of the full-set battery starts a greedy backward elimination (`ils::selectFixableSubset` in `src/core/filter/IlsCore.cpp`, called through the driver-side `selectFixableSubset_` configuration adapter) before the epoch falls back to float. A failure of MLAMBDA itself on the full set (a covariance that is not positive definite, or fewer than two candidates) does not start the elimination. Starting from the full candidate set, each iteration re-runs the complete MLAMBDA pipeline of Sections 1.1 to 1.2 on the current subset and re-tests the same acceptance battery: the ratio lower and upper bounds (15), the bootstrap success rate (16), and the per-entry fractional cap (17). On any failure it drops one entry and retries, down to a single ambiguity:

- if the *fractional cap* is the failing gate, the entry with the largest $|\hat{\boldsymbol{N}}_k - \check{\boldsymbol{N}}_k|$ (nearest a half-cycle boundary) is removed;
- otherwise (including a failure of MLAMBDA on the subset) the entry with the largest formal float variance $(\boldsymbol{Q}_{\hat{N}})_{kk}$ is removed: the least precise ambiguity, whose removal is expected to improve the bootstrap rate and the search conditioning most.

The accepted subset is always re-validated by the full battery on its own covariance sub-block. The procedure can therefore only fix fewer ambiguities than the full set, never accept a set that fails the battery: a poor elimination order costs fixes, not integrity. With the option off, the path is bit-identical to the all-or-nothing behaviour. The same routine serves the per-band SD, baseline DD and WL-constrained NL paths and operates only in the cycle domain, independent of the decorrelation. Each caller maps the surviving indices back to its own state rows and applies them through the constrained update of Section 1.6.

1.5 Validation: ratio test and fixed-failure-rate gate

A fix is committed only if it passes all three of the following gates. The code evaluates them in order (ratio lower bound, ratio upper bound, bootstrap success rate, per-pair cap) and stops at the first failure. The ratio and fixed-failure-rate (FFR) gates bound different things: the ratio bounds the discrimination between the two best candidates, the FFR gate the absolute success probability.

Ratio test (discrimination).

$$\boxed{\text{ratio} = \frac{s_2}{s_1} \;\geq\; \tau_{\text{ratio}}, \qquad s_1 \leq s_2 \text{ the two smallest ILS objectives,}} \tag{15}$$

with s_1, s_2 from the search of Section 1.2 and threshold τ_{ratio} (`ratio_threshold`). An upper-bound rejection, ratio $> \tau_{\text{ratio}}^{\max}$ (`ratio_upper_bound`), catches a degenerate covariance direction: a near-zero s_1, typically left by an earlier wrong-integer hold, produces a very large but meaningless ratio. A non-positive squared distance for either candidate rejects the fix before the ratio is formed.

Fixed-failure-rate gate (bootstrap success rate). The gate uses a lower bound on the ILS success probability that follows from the problem geometry,

$$\boxed{P_{s,\mathrm{boot}} = \prod_{i=1}^{n} \operatorname{erf}\!\left(\frac{1}{2\sqrt{2\,d_i}}\right) \;\geq\; P_s^{\min},} \tag{16}$$

where the d_i are the post-reduction LDL^{T} conditional variances of (5) and $P_s^{\min}$ is `min_success_rate`. Each factor is $\Pr\big(|\mathcal{N}(0,d_i)| < \frac{1}{2}\big)$, the probability that sequential bootstrapped rounding lands on the correct integer in that dimension. The product is Teunissen's lower bound on the ILS success rate. The acceptance threshold is therefore tied to the problem geometry rather than to a fixed ratio alone.

Per-pair residual backstop. The joint ratio test can pass on a borderline set in which one pair lies near a half-cycle boundary. The fix is therefore rejected if the fractional residual of any single DD exceeds a cap,

$$\max_k \big|\hat{\boldsymbol{N}}_k - \check{\boldsymbol{N}}_k\big| \;\leq\; \kappa_{\mathrm{frac}} \quad \text{(cycles)}, \tag{17}$$

with $\kappa_{\mathrm{frac}} =$ `nl_max_per_pair_frac`.

Convergence guards. Before any search the resolver skips AR while the float solution is immature: during its first `min_epochs` invocations (a global counter of resolver calls, not a per-candidate age), and while the position standard deviation $\sqrt{\boldsymbol{P}_{00}+\boldsymbol{P}_{11}+\boldsymbol{P}_{22}}$ exceeds `max_position_sigma_m`. A marginal fix accepted before the covariance settles can lock onto wrong integers and corrupt every later epoch.

1.6 Feedback: constrained pseudo-observation update

Accepted integers are fed back through a finite-variance pseudo-observation rather than the classical perfect back-substitution (the $R = 0$ conditioning form)

$$\boldsymbol{x}_{\mathrm{fix}} = \boldsymbol{x} - \boldsymbol{Q}_{\boldsymbol{aa}}\,\boldsymbol{Q}^{\nabla\Delta-1}\big(y^{\nabla\Delta} - \check{\boldsymbol{N}}\big), \qquad \boldsymbol{P}_{\mathrm{fix}} = \boldsymbol{P} - \boldsymbol{Q}_{\boldsymbol{aa}}\,\boldsymbol{Q}^{\nabla\Delta-1}\boldsymbol{Q}_{\boldsymbol{aa}}^{\mathrm{T}}, \tag{18}$$

The update is a *Joseph-form pseudo-observation* (`src/core/filter/AmbiguityResolver.cpp`; kernel `KalmanFilter::josephPseudoUpdate` in `include/filter/KalmanFilter.hpp`). The reason is structural. (18) drives the fixed ambiguity rows of $\boldsymbol{P}$ to ≈ 0 variance, so the next epoch's LAMBDA sees a near-singular $\boldsymbol{Q}^{\nabla\Delta}$ and returns large, meaningless ratios that keep tripping the upper-bound guard of (15). A finite-variance pseudo-observation is intended to keep $\boldsymbol{P}$ well-conditioned, so that later epochs can re-confirm the integer (converging toward it) or detect a slip through its residual.

This intent is not yet validated. Reducing σ^2_{hold} (over the range 1e-3 to 1e-7, in square metres) reduced the fix count and made the kinematic solution diverge, which is the signature of a constraint that drives the covariance singular rather than tightening it. The resolver therefore carries a read-only conditioning probe (`qbCondition` / `rebuildQbCycles` in `src/core/filter/AmbiguityResolver.cpp`) that eigen-decomposes the cycle-domain DD covariance before every search and after every fix, and logs the conditioning on every fix and every rejection. The mechanism described below is the design intent, not a validated result.

For each accepted DD pair k the constraint row, innovation and noise are

$$\boxed{\boldsymbol{H}_{(k,\,i)} = +1, \quad \boldsymbol{H}_{(k,\,p)} = -1, \qquad v_k = \lambda_k\,\check{\boldsymbol{N}}_k - \big(\boldsymbol{x}_i - \boldsymbol{x}_p\big), \qquad \boldsymbol{R} = \sigma^2_{\mathrm{hold}}\,\boldsymbol{I},} \tag{19}$$

followed by $(\boldsymbol{x}_{\text{fix}}, \boldsymbol{P}_{\text{fix}}) = \textsc{JosephUpdate}(\boldsymbol{x}, \boldsymbol{P}, \boldsymbol{H}, \boldsymbol{v}, \boldsymbol{R})$, with $\sigma^2_{\text{hold}} =$ `hold_variance`. The DD/baseline path uses a single-column selector $\boldsymbol{H}_{(k,\,\text{idx})} = +1$, because the state value already is the double difference. The same update applies a partial-AR subset, with fewer rows. The updated state and covariance replace the filter's on every epoch with an accepted fix, in every mode other than 0. The resolver does not distinguish modes 1 (continuous) and 2 (instantaneous); only mode 0 (float) and mode 3 (fix-and-hold) change its behaviour. The $R = 0$ form (18) is not implemented; it appears in the code only in comments, as the rejected alternative.

Fix-and-hold mode (`estimation.ambiguity_resolution.mode` = 3) applies a second pseudo-observation update on each epoch with an accepted fix, holding the differenced ambiguities at their fixed values with the same hold variance. In the baseline path (AmbiguityResolver::holdAmbiguitiesDd) the hold covers every DD-keyed ambiguity in the state, including keys excluded from the fix. In the single-receiver path (AmbiguityResolver::holdAmbiguities, called by both the post-processing and the real-time engines) the hold is inert under the default configuration. It rebuilds the DD pairs with an empty observation list, so every ambiguity is assigned elevation 0 and the AR elevation mask (default 20°) removes every pair. PPP fix-and-hold therefore holds nothing unless the mask is 0 or negative; mode 3 then behaves like modes 1 and 2. Under the NL/WL state basis the hold is a no-op by design.

1.7 Network-producer AR (both mechanisms default off)

The PPP-RTK network producer of the Network mode document is float-only. Its per-epoch seam sequence contains two integer mechanisms, both default-off; with the defaults, neither changes the emitted satellite-bias product. They are documented here because their algebra is settled, not because they are in use.

Producer AR: a log-only fixability probe. Under `network.ambiguity_resolution.enable` (default `false`) the seam `filter/NetworkProducerAr` reconstructs each station's float ambiguities from the reduced solution (`NetworkEstimator::reconstructStation`) and forms an independent between-station double-difference basis: a star topology per (constellation, signal), restricted to GPS and Galileo because GLONASS FDMA leaves a per-slot receiver term that survives the difference. It then builds the exact DD covariance from `NetworkEstimator::jointConditionalCovariance`, rescales it to cycles by (4), and calls `ils::ilsFix`. The result is logged and discarded: nothing is re-pinned or re-solved, and the run is byte-identical whether the block executes or not. The probe is meaningful because the between-station DD ambiguity

$$\nabla\Delta N^{i,p_g}_{r_1 r_2,s} \;=\; \Delta N^{i,p_g}_{r_1,s} - \Delta N^{i,p_g}_{r_2,s} \tag{20}$$

is invariant under the producer's choice of phase-bias S-basis: the satellite bias b^i_s cancels in the between-receiver difference exactly as it does in the baseline DD of the double-difference section of the Foundations document. The probe therefore reports the network's intrinsic integer recoverability rather than a property of the datum the producer holds. Its convergence gauge is the median DD formal standard deviation in cycles. That value must fall well below the half-cycle scale before (1) is reachable, and it is a better progress indicator than the fix count. Two configuration notes apply. `ils_max_dim` (default 0 = uncapped) exists because the search does not return on the full-network basis at operational station counts; a positive value restricts the search to the lowest-formal-σ subset, in the manner of Section 1.4. `wl_cascade` is parsed and echoed in the load-time log but has no consumer in the code, so it is inert.

SD extra-wide-lane integers: Route B and the A2 datum. The second mechanism (`filter/NetworkSdEwlAr`, gated on `network.sd_ewl.fix_enable`, default `false`) decides an integer for the between-satellite single-differenced extra-wide-lane ambiguity. The motivation is a rank argument. The float Melbourne–Wübbena row that the enclosing SD-EWL layer folds is a linear combination of observables already in the epoch's normal equations, so it adds no rank; the only new information the row can supply is integrality. Rounding the SD statistic directly is circular, because it requires the satellite bias difference $b^{i}_{\mathrm{EWL}} - b^{p_g}_{\mathrm{EWL}}$ that the network estimates. Route B avoids the circularity by differencing the SD MW statistic between stations, which gives the same double difference as (20):

$$\boxed{\left(\mathrm{MW}^{i}_{r_1} - \mathrm{MW}^{p_g}_{r_1}\right) - \left(\mathrm{MW}^{i}_{r_2} - \mathrm{MW}^{p_g}_{r_2}\right) \;=\; \lambda_{\mathrm{EWL}}\,\nabla\Delta N^{i,p_g}_{r_1 r_2,\mathrm{EWL}} \;+\; \nabla\Delta\varepsilon^{\mathrm{MW}},} \tag{21}$$

in which the satellite bias pair cancels exactly: the satellite pair is the same on both sides, and each station's receiver term has already cancelled inside its own single difference. What remains is an integer plus averaged MW noise at the extra-wide-lane wavelength. An integer decided this way uses no a-priori satellite bias and cannot be corrupted by a wrong one, which is why Route B is preferred over rounding against a bias product. The cancellation in (21) has a precondition that belongs to the algebra, not to the implementation: b_{EWL} is a per-signal combination, so both stations' arcs must use the same two tracking channels. Otherwise a per-signal bias difference is folded in as though it were an integer. The seam therefore elects a network-common channel pair per satellite by station consensus and admits only stations that hold that pair.

Equation (21) yields a double difference, while the constrained row needs a single-differenced integer per (station, satellite). The A2 datum supplies one. Within each connected component a reference station r_0 is elected, $\Delta n^{i,p_g}_{r_0} := 0$ is defined, and the rounded DD fixes are propagated outward along a spanning tree; the link satellite absorbs the per-station free integer. The asserted quantity is then

$$\Delta n^{i,p_g}_{r} \;=\; \Delta N^{i,p_g}_{r,\mathrm{EWL}} - \Delta N^{i,p_g}_{r_0,\mathrm{EWL}}, \tag{22}$$

the true SD integer minus a constant per (satellite pair, component), which is the classical wide-lane fractional-cycle-bias convention modulo λ_{EWL} [1, 2]. The constant is absorbed into the estimated bias the network reports. Two consequences of (22) matter for the design. First, the per-station free integer, and with it the spanning tree, cancels at the point of use, so a change of spanning tree between epochs is not a defect. Second, r_0's own integers are absorbed into the constant, so the choice of r_0 is a datum decision rather than a measurement.

Acceptance is the scalar case of the battery in Section 1.5. A rounded DD is admitted on its fractional residual, as in (17), and on the $n = 1$ instance of the bootstrap bound (16), each with its own threshold (network.sd_ewl.fix_max_frac_cyc, default 0.15 cycle, and network.sd_ewl.fix_min_ffr, default 0.999):

$$P_{s,\mathrm{round}} = \mathrm{erf}\left(\frac{1}{2\sqrt{2}\,\sigma}\right), \qquad \sigma \text{ in cycles}, \tag{23}$$

so a single rounding and the n-dimensional search share the same failure-rate semantics.

The mechanism is subject to the standing rule that a stiff, time-linked quantity needs a wired event response. The fixed row writes a λ_{EWL}-quantised value into satellite bias states with very small process noise, so an integer that changes between epochs would inject a step that those states take hours to relax out of. The wired response is withhold-on-disagreement. When this epoch's integer differs from the previous one on a continuous arc, the fix for that arc is withheld rather than repaired, because on a disagreement epoch neither value is known to be right. An arc that restarted (slip, gap, re-acquisition) replaces the stored integer without counting as a disagreement, because its

integer has legitimately changed. The stored integer is used only to veto, never to supply a value; this keeps the mechanism short of a carried zero-process-noise hold, which is deliberately not built.

Status. Both mechanisms are default-off, and neither has a graded, datum-invariant field result. The loader also disarms `network.sd_ewl.fix_enable`, with a warning, unless the float row layer (`network.sd_ewl.enable`) is armed. That layer is itself inert without `network.phase_leveling` and an MW-class slip detector, because it reads the arc accumulator they build. The emitted network product is float.

1.8 Configuration summary

All tunable thresholds live under the JSON path `estimation.ambiguity_resolution.*`, parsed in `src/core/config/GnssConfigLoad.cpp` with defaults in `include/filter/AmbiguityResolver.hpp`. The estimator theory of Sections 1.1 to 1.6 has no settings, since changing it would change the mathematics rather than a policy. In the mode row of the table, modes 1 and 2 behave identically in the code (Section 1.6). The table omits three WL acceptance keys described in Section 1.3 (wl_only_ffr, default on; wl_only_min_arc_count, default 10; wl_only_max_innov_cyc, default 0.5) and the search-row option nl_iono_free_row (default off). The two network-producer mechanisms of Section 1.7 are configured separately, under `network.ambiguity_resolution.*` and `network.sd_ewl.*` (parsed in `src/core/config/GnssConfigLoadNetwork.cpp`); both master switches default to `false` and are omitted from the table.

Symbol	Config key (`...ambiguity_resolution.`)	Default	Role
	`mode`	`0`	0 float / 1 continuous / 2 instantaneous / 3 fix-and-hold
τ_{ratio}	`ratio_threshold`	`3.0`	ratio-test acceptance (15)
$\tau_{\text{ratio}}^{\max}$	`ratio_upper_bound`	`1e4`	degenerate-covariance reject (15)
$P_s^{\min}$	`min_success_rate`	`0.999`	FFR gate (16)
κ_{frac}	`nl_max_per_pair_frac`	`0.15`	per-pair residual cap (17)
θ_{AR}	`elevation_mask`	`20°`	candidate elevation mask
σ_{hold}^2	`hold_variance`	`0.001` m^2	pseudo-obs $\boldsymbol{R}$ (19)
$n_{\min}^{\text{ep}}$	`min_epochs`	`10`	convergence guard
$\sigma_{\text{pos}}^{\max}$	`max_position_sigma_m`	`0.05` m	convergence guard
$\sum \sin^2\theta_{\min}$	`min_arc_info`	`0.0` (off)	per-candidate arc-maturity gate, WL/NL path only (Section 1.4)
	`wl_only`	`false`	MW-WL-only, no NL search
σ_{wl}^2	`wl_only_pseudo_var`	`1e-4` cyc^2	WL-constraint pseudo-obs $\boldsymbol{R}$ (Section 1.3)
	`wl_max_frac`	`0.40`	per-arc WL fractional-residual gate (Section 1.3)
	`enable_partial_ar`	`false`	backward-elimination subset fix (Section 1.4)
	`active_systems`	`[]` (all)	constellation subset (Section 1.4)
	`state_ambiguity_basis` †	`per_band`	`per_band` vs `nl_wl` dispatch

† Path is `estimation.state_ambiguity_basis`, one level above the `ambiguity_resolution` block.

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